

Evaluation of steel tape armour and copper screening in railway signaling cables: impact on screening factor

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Abstract—In today's world, as the power and sensitivity of electrical systems increase, addressing Electromagnetic Compatibility (EMC) becomes crucial. One such area of focus is electric traction and the signaling cables used for its safety devices. In our study, we examined low-voltage railway signaling cables. These cables are designed to transmit essential data and information about the railway system, as well as to operate safety devices along the railway. Therefore, their precise and reliable functioning is critical. However, without proper awareness, higher voltages may be induced in these cables, posing risks. Voltage reduction can be achieved by shielded cables, surrounding the transmission cable's wires with a conductive layer. In this layer, current is generated by the continuously changing magnetic field. According to Lenz's law, this current opposes the magnetic field (flowing in the opposite direction to the inducing current), thus weakening the magnetic field inside the layer compared to outside. This effectiveness is quantified by the screening factor. In our measurements, we explored the dependency of the screening factor on the different pieces of the cable. What role does steel tape armor play? What role does the copper layer serve? This topic remains highly relevant today, especially with the increasing traction currents appearing globally. Our study presents new findings that aim to clarify certain issues and propose new solutions.

Keywords - signalling cable, screening factor, steel tape armour, copper layer

I. INTRODUCTION

When it comes to Electromagnetic Compatibility (EMC) in railway systems, the main issue arises from the alternating high traction current. As a result, high voltage is induced in the signaling cables, leading to serious consequences. [1] [2] [3]

To protect these crucial signaling cables, various solutions exist. This paper focuses on the use of shielded cables. These cables are typically multi-core cables, shielded by a copper layer and steel tape armoring. They are generally manufactured in 500 m lengths; however, railway lines are much longer. To address this issue, the 500 m long sections should be connected. In order to preserve the original shielding effect (screening factor), ensuring proper connections between these cable sections is essential. [4]

For a reliable connection, it is crucial to join the copper layers and steel tape armor with the minimal contact resistance possible. If the resistance is high, the connection can be modeled as a break, which can lead to an inadequate screening factor.

In our study, we modeled two scenarios in the laboratory where one of the two layers was not perfectly connected. We measured the screening factor for both configurations and compared the results to those of the original setup, which represents the properly connected case.

II. METHOD OF MEASUREMENTS

A. Measurement of screening factor

The measuring environment was built according to [5], and it is shown in Fig. 1.

To calculate the inner screening factor of the cable, U_k outer surface voltage was measured between the points k_p and k , with the use of an auxiliary outer surface conductor. The U_b inner surface voltage was measured between the points k and b , using a cable conductor. For the measurement of the currents I , current probes were used. These work similarly to current transformers, but instead, they provide voltage that is proportional to the induction's differential quotient. With the use of the measured surface voltages and the resultant current, the Z_k outer and Z_b inner surface impedances can be calculated by definitions (1) [5]

$$|Z_k| = \frac{|U_k|}{|I|} \quad |Z_b| = \frac{|U_b|}{|I|} \quad (1)$$

The inner screening factor can be calculated according to (2). Thus the screening factor is a number between 0 and 1, and the smaller its value the better the screening effect is.

$$|k_b| = \frac{|U_b|}{|U_k|} = \frac{|Z_b|}{|Z_k|} [-] \quad (2)$$

From (2) it can be concluded that the inner screening factor is purely cable-specific. It does not depend on the soil resistivity

of the installation location or the earthing conditions. Therefore, the inner screening factor is suitable to compare different cables' screening effectiveness. [5]

During the measurements, the variable value was the current I , which was varied from 0 to 150 A in steps of 10 A. Using this technique, the inner screening factor can be calculated for different current values.

B. Modelling the inappropriately connected configurations

When the continuity of the copper layer or the steel tape armour is not adequate, in the worst case scenario, it can be modeled as a break, meaning that the current cannot flow through that layer. To simulate this situation, in the first measurement, the steel tape armour (at point "A" in Fig. 1), and during the second measurement, the copper layer (at point "B" in Fig. 1) were intentionally disconnected from the loop. As a result, the current flowed only through one part of the cable.

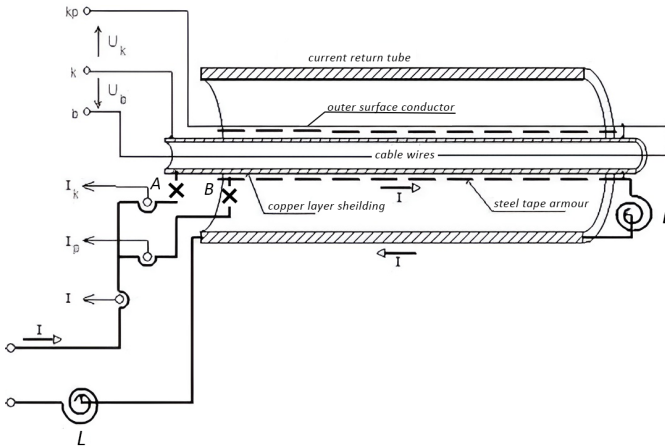


Fig. 1 — Measurement setup

After these modifications in the measurement setup, the screening factor was measured in the same way as described above.

C. The measurement technique

To measure the currents and the surface voltages, an oscilloscope was used. The most challenging part was the measurement of the inner surface voltage due to its small magnitude. To address this, an oscilloscope with 16-bit resolution was selected, providing higher quantization precision. However, since resolution alone does not ensure accurate detection of low-level signals—especially in the presence of noise—additional steps were taken to minimize external interference. Proper grounding and differential probes were used to reduce noise and enhance signal integrity.

At one measurement point (i.e., 30 A), three periods of the waveform were captured, and the surface voltage and current values were calculated from these. These measures, combined with high-resolution acquisition, significantly improved the accuracy of the measurements.

III. COMPARING THE ROLE OF STEEL TAPE ARMOUR AND COPPER LAYER

A. Inner screening factor

In the third part of this paper, the effect of the steel tape armour and copper layer on the forming of the screening factor is examined. To begin, the results of the inner screening factor measurements are presented in Fig. 2.

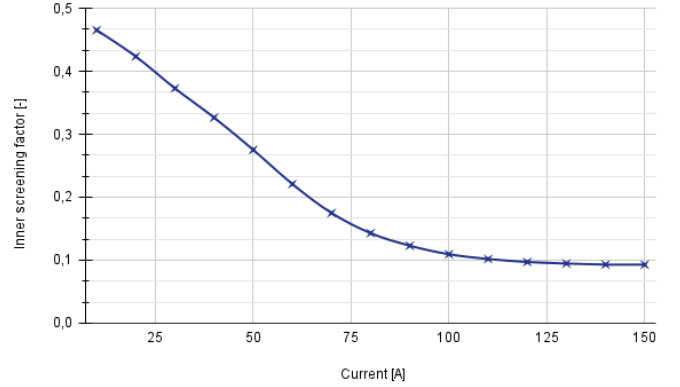


Fig. 2 — Screening factor of the cable in function of current

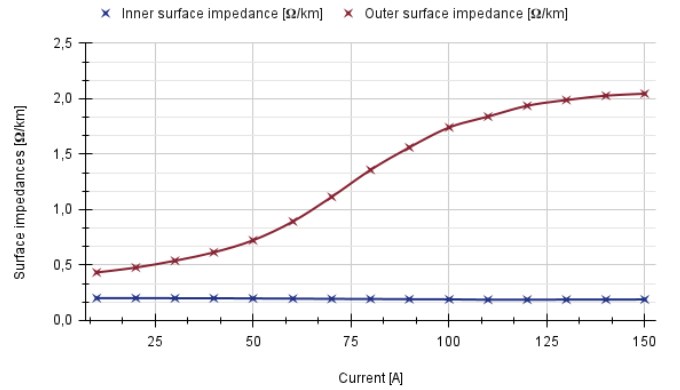


Fig. 3 — Surface impedance values of the cable in function of current

As it is shown in (1) the inner screening factor correlates with the ratio of inner and outer surface impedances. The results are presented in Fig. 3

During our measurements, the same signaling cable was used.

TABLE I: Tested Cable Details

Type	Measured length [m]
Railway line cable	3.1

B. Role of steel tape armour

For this measurement the I_o should be 0. In order to achieve this, the steel tape armour was disconnected from the loop with the connector "B", Fig. 1. With this disconnection,

a scenario was modeled when the connection of steel tape armours between two cable sections is inappropriate.

The results of surface impedances in Fig. 4, and inner screening factors are presented in Fig. 5.

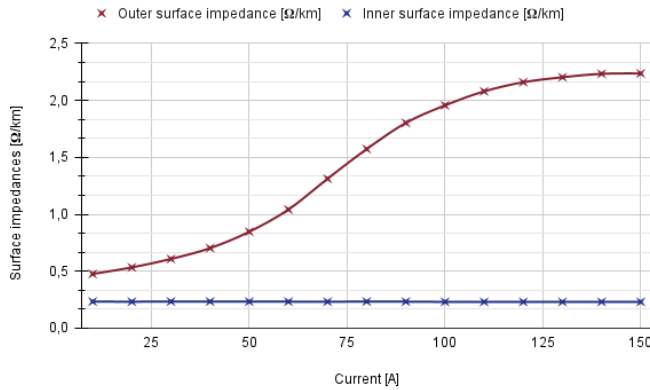


Fig. 4 — Surface impedance values - copper wire screening

In Fig. 4 the values of outer surface impedances are nearly identical to those in Fig. 3. Furthermore, only small changes can be observed in the values of inner surface impedance.

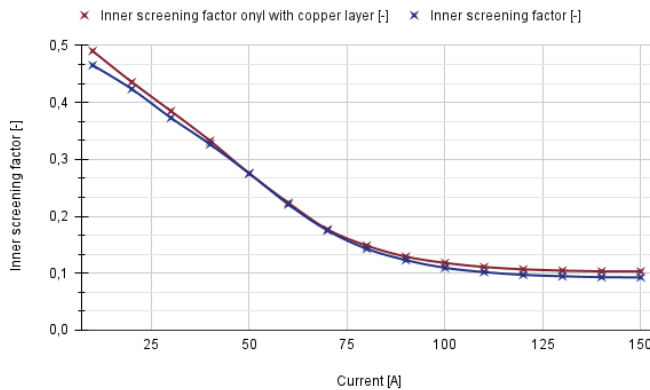


Fig. 5 — Inner screening factor values - copper wire screening

Because of the similar impedance values, the values of the inner screening factor also stayed quite similar to those in Fig. 2. The only mentionable difference is the slightly worse inner screening factor, thanks to the reduced conduction ability of the cable, since the steel tape armour was disconnected from the circuit.

C. Role of copper layer

For this measurement the I_{armour} should be 0, which represents the scenario when the connection of the copper layer between the cables is not appropriate. In order to achieve this, the copper layer was disconnected from the loop at point "A" on Fig. 1.

The results of surface impedances in Fig. 6, and inner screening factors can be seen in Fig. 7.

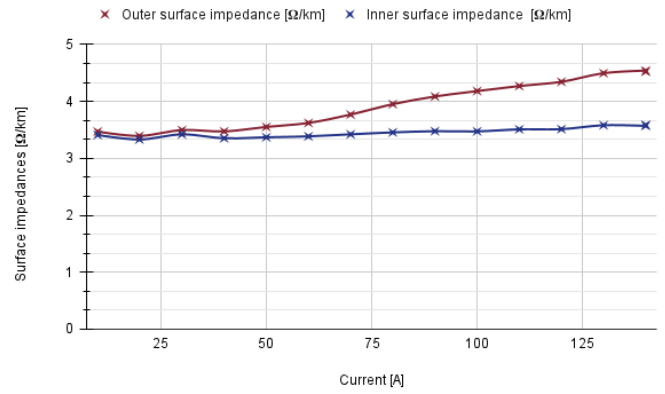


Fig. 6 — Surface impedance values - steel tape armour

Disconnecting the copper layer from the circuit has increased the values of the inner and outer surface impedances, as it can be observed in Fig. 6. Another noticeable difference shown in the graph is that the two impedances are nearly identical at low current levels, with their values remaining closely aligned at higher current levels. This has a direct effect on the inner screening factor of the cable, Fig. 7

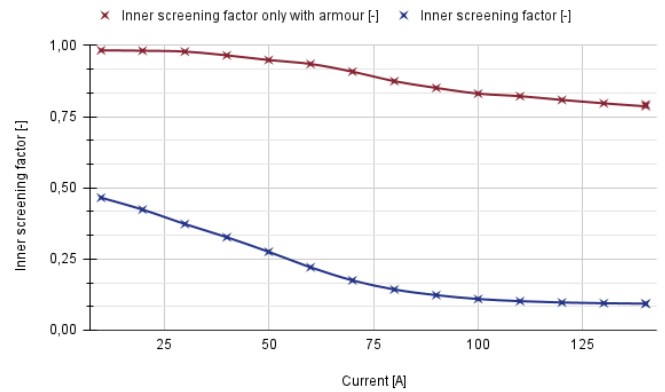


Fig. 7 — Inner screening factor values - steel tape armour

In this case, as shown in Fig. 7, the value of the inner screening factor has drastically increased. At low current levels, it is nearly one, which means no protection is provided against the induced effect. At higher current levels, the inner screening factor is still quite far from the original screening factor.

D. Conclusion

The aim of this paper is to provide practical results and recommendations for the railway industry regarding the electromagnetic compatibility of railway signalling cables. Based on the results presented above, it is evident that improper connection of the steel tape armours between two cable sections does not significantly reduce the cable's internal screening factor.

In terms of copper layer continuity, the situation is slightly different. As it is demonstrated above, the inappropriate connection of the copper layers leads to serious consequences and immensely reduces the inner screening factor of the cable. Therefore, in practice, it is crucial to perfectly connect the copper layers.

This investigation is being continued in order to find and reveal the causes of the inner screening factor deterioration.

REFERENCES

- [1] Donald G. Baker. *Electromagnetic Compatibility: Analysis and Case Studies in Transportation*. Hoboken, NJ: Wiley, 2017.
- [2] Dieter Stoll. *EMC elektromágneses zavarvédelem*. Műszaki Könyvkiadó, 1980.
- [3] Author(s) Unknown. "Analysis of Induced Voltage at Close Proximity with High Voltage Cables of Electrified Railways". In: *Proceedings of the IEEE International Symposium on Electromagnetic Compatibility (EMC Europe)*. Graz, Austria, 2018.
- [4] F. Kiessling et al. *Contact Lines for Electric Railways: Planning, Design, Implementation, Maintenance*. 3rd. Erlangen, Germany: Publicis Publishing, 2018. ISBN: 978-3-89578-443-5.
- [5] Jozsef Ladanyi. "Examination of screening factor of signaling cables by measurements and calculation". In: *2017 International Symposium on Electromagnetic Compatibility - EMC EUROPE*. 2017, pp. 1–6. DOI: 10.1109/EMCEurope.2017.8094721.
- [6] International Telecommunication Union. *ITU - CCITT - DIRECTIVES - Volume IX*. 1989.
- [7] N. Karimian et al. "Differential permeability behaviour of P9 and T22 power station Steels". In: *Journal of Magnetism and Magnetic Materials* 352 (2014), pp. 81–90. ISSN: 0304-8853. DOI: <https://doi.org/10.1016/j.jmmm.2013.09.059>. URL: <https://www.sciencedirect.com/science/article/pii/S030488531300718X>.
- [8] Matthew N. O. Sadiku. *Elements of Electromagnetics*. 6th. Oxford, UK: Oxford University Press, 2015.